

MAE 6246 - Linear Systems and Control

Week 3: Solutions of Continuous-Time Linear Systems

Fall 2026

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Topics: homogeneous and forced responses; zero-input and zero-state responses; matrix exponential; Laplace resolvent; fundamental matrices; state-transition matrices

1 Learning objectives

By the end of the lecture, you should be able to:

1. derive the solution of an LTI initial-value problem;
2. separate a response into zero-input and zero-state components;
3. state and use the defining properties of the matrix exponential;
4. connect the time-domain solution with the Laplace resolvent $(sI - A)^{-1}$;
5. distinguish a fundamental matrix from the state-transition matrix;
6. state the identity, inverse, composition, and differential properties of a state-transition matrix.

The central message is that the state at time t is the sum of the propagated initial condition and the accumulated effect of the input. The state-transition matrix is the object that performs both operations.

2 Continuous-time linear systems

A linear time-varying (LTV) state model has the form

$$\dot{x}(t) = A(t)x(t) + B(t)u(t), \quad y(t) = C(t)x(t) + D(t)u(t),$$

where the matrices may depend explicitly on time. A linear time-invariant (LTI) model is the special case

$$\boxed{\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t) + Du(t),}$$

with constant matrices.

For a specified initial time t_0 , initial state $x(t_0) = x_0$, and input $u(t)$, the problem is to determine the unique trajectory $x(t)$ satisfying the differential equation and the initial condition.

The adjective *linear* refers to the dependence on x and u . Time variation alone does not make a system nonlinear. Nevertheless, formulas that rely on a constant matrix A cannot be transferred automatically to the LTV case.

3 Homogeneous LTI response and the matrix exponential

First set the input to zero:

$$\dot{x} = Ax, \quad x(t_0) = x_0.$$

The scalar equation $\dot{z} = az$ has solution $z(t) = e^{a(t-t_0)}z(t_0)$. Its matrix counterpart is

$$\boxed{x(t) = e^{A(t-t_0)}x_0.}$$

The matrix exponential is defined by the convergent power series

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots$$

Term-by-term differentiation gives

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A,$$

and therefore $x(t) = e^{A(t-t_0)}x_0$ satisfies both the differential equation and the initial condition.

3.1 Basic properties

For a fixed matrix A ,

$$e^{A0} = I, \quad e^{A(t_2-t_0)} = e^{A(t_2-t_1)}e^{A(t_1-t_0)}, \quad (e^{At})^{-1} = e^{-At}.$$

In general,

$$e^{(A+B)t} \neq e^{At}e^{Bt}.$$

The equality holds when A and B commute, meaning $AB = BA$.

3.2 Computing the matrix exponential from structure

If $A = PAP^{-1}$ is diagonalizable, then

$$e^{At} = Pe^{\Lambda t}P^{-1}, \quad e^{\Lambda t} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

For a Jordan block $J = \lambda I + N$, where $N^m = 0$,

$$e^{Jt} = e^{\lambda t} \left(I + Nt + \frac{N^2 t^2}{2!} + \dots + \frac{N^{m-1} t^{m-1}}{(m-1)!} \right).$$

Thus defective modes contain polynomial factors multiplying $e^{\lambda t}$. In numerical work, use a stable matrix-exponential routine such as **scipy.linalg.expm**; explicit Jordan-form computation is primarily a conceptual tool.

Example: a 3×3 Jordan block. Consider

$$J = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix} = \lambda I + N, \quad N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The nilpotent part satisfies

$$N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad N^3 = 0.$$

Because λI commutes with N ,

$$e^{Jt} = e^{(\lambda I + N)t} = e^{\lambda t} e^{Nt}.$$

The exponential series for N terminates after the quadratic term:

$$e^{Nt} = I + tN + \frac{t^2}{2}N^2.$$

Therefore,

$$e^{Jt} = e^{\lambda t} \begin{bmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}.$$

The entries on successive superdiagonals contain successively higher powers of t .

3.2.1 Diagonalizable example from Week 2

Consider

$$A = \begin{bmatrix} 3 & -2 \\ -1 & 4 \end{bmatrix}, \quad P = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}.$$

As obtained in Week 2,

$$\Lambda = P^{-1}AP, \quad A = P\Lambda P^{-1},$$

where

$$P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}.$$

The power-series definition of the exponential gives

$$e^{At} = P e^{\Lambda t} P^{-1} = P \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{5t} \end{bmatrix} P^{-1}.$$

Multiplying the three matrices yields

$$e^{At} = \frac{1}{3} \begin{bmatrix} 2e^{2t} + e^{5t} & 2e^{2t} - 2e^{5t} \\ e^{2t} - e^{5t} & e^{2t} + 2e^{5t} \end{bmatrix}.$$

At $t = 0$, this matrix is I . Differentiating it and evaluating at $t = 0$ gives A , providing two quick checks of the calculation.

3.2.2 Defective example from Week 2

Now consider

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}.$$

Its characteristic polynomial is $(\lambda - 2)^2$. An eigenvector and a generalized eigenvector can be chosen as

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

because

$$(A - 2I)v_1 = 0, \quad (A - 2I)v_2 = v_1.$$

With

$$P = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix},$$

the Jordan form is

$$J = P^{-1}AP = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = 2I + N, \quad N^2 = 0.$$

Therefore,

$$e^{Jt} = e^{2t}e^{Nt} = e^{2t}(I + tN) = e^{2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix},$$

and

$$e^{At} = Pe^{Jt}P^{-1} = e^{2t} \begin{bmatrix} 1+t & t \\ -t & 1-t \end{bmatrix}.$$

Equivalently, $A = 2I + (A - 2I)$ and $(A - 2I)^2 = 0$, so the same result follows directly from

$$e^{At} = e^{2t}[I + t(A - 2I)].$$

4 Forced LTI response

Return to

$$\dot{x} = Ax + Bu.$$

Multiply by e^{-At} and use the product rule:

$$\frac{d}{dt}(e^{-At}x(t)) = e^{-At}(\dot{x} - Ax) = e^{-At}Bu(t).$$

Integrating from t_0 to t gives

$$e^{-At}x(t) - e^{-At_0}x(t_0) = \int_{t_0}^t e^{-A\tau}Bu(\tau) d\tau.$$

Multiplying by e^{At} yields the variation-of-constants formula

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau) d\tau.$$

The two terms have different physical meanings.

4.1 Zero-input and zero-state responses

The **zero-input response** is caused only by the initial condition:

$$x_{\text{zi}}(t) = e^{A(t-t_0)}x_0.$$

The **zero-state response** is caused only by the input:

$$x_{\text{zs}}(t) = \int_{t_0}^t e^{A(t-\tau)}Bu(\tau) d\tau.$$

By linearity,

$$x(t) = x_{\text{zi}}(t) + x_{\text{zs}}(t).$$

The output follows directly:

$$y(t) = Ce^{A(t-t_0)}x_0 + \int_{t_0}^t Ce^{A(t-\tau)}Bu(\tau) d\tau + Du(t).$$

The kernel

$$G(t) = Ce^{At}B, \quad t \geq 0,$$

is the strictly proper part of the impulse response. The term $Du(t)$ is the direct feedthrough.

5 Laplace-transform and resolvent derivation

Let $t_0 = 0$ and denote the unilateral Laplace transforms by $X(s)$ and $U(s)$. Since

$$\mathcal{L}\{\dot{x}\} = sX(s) - x(0),$$

the state equation gives

$$sX(s) - x(0) = AX(s) + BU(s).$$

Therefore,

$$(sI - A)X(s) = x(0) + BU(s),$$

and

$$\boxed{X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s).}$$

The matrix

$$\boxed{R(s; A) = (sI - A)^{-1}}$$

is the **resolvent** of A . Its poles occur at the eigenvalues of A because

$$\det(sI - A) = 0$$

at those values. The identity

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1}$$

connects the time-domain matrix exponential to the frequency-domain resolvent.

5.1 Convolution theorem

Let

$$F(s) = \mathcal{L}\{f(t)\}, \quad G(s) = \mathcal{L}\{g(t)\}.$$

The convolution theorem states that

$$\boxed{\mathcal{L}\{(f * g)(t)\} = F(s)G(s),}$$

where

$$(f * g)(t) = \int_0^t f(t - \tau)g(\tau) d\tau.$$

Equivalently,

$$\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(t - \tau)g(\tau) d\tau.$$

Apply this identity to the forced term in $X(s)$. Since

$$\mathcal{L}^{-1}\{(sI - A)^{-1}B\} = e^{At}B,$$

we obtain

$$\mathcal{L}^{-1}\{(sI - A)^{-1}BU(s)\} = \int_0^t e^{A(t-\tau)}Bu(\tau) d\tau.$$

Thus, the product in the Laplace domain becomes the zero-state convolution integral in the time domain.

For zero initial condition, the transfer matrix is

$$G(s) = C(sI - A)^{-1}B + D.$$

The transfer matrix describes input-output behavior, whereas the state-transition solution retains the internal state and the effect of the initial condition.

6 Fundamental matrices for homogeneous LTV systems

Consider the homogeneous LTV system

$$\dot{x} = A(t)x.$$

The set of its solutions is a linear space: any linear combination of solutions is again a solution. Choose n linearly independent solutions

$$\psi_1(t), \dots, \psi_n(t)$$

and place them into the columns of

$$\Psi(t) = [\psi_1(t) \quad \cdots \quad \psi_n(t)].$$

Then

$$\dot{\Psi}(t) = A(t)\Psi(t).$$

If $\Psi(t)$ is nonsingular, it is a **fundamental matrix**. A fundamental matrix is not unique: multiplying it on the right by any constant nonsingular matrix produces another fundamental matrix.

7 State-transition matrix

Given any fundamental matrix $\Psi(t)$, define

$$\Phi(t, t_0) = \Psi(t)\Psi^{-1}(t_0).$$

Although Ψ is not unique, $\Phi(t, t_0)$ is unique for a given $A(t)$. It maps a state at t_0 to the corresponding homogeneous state at t :

$$x(t) = \Phi(t, t_0)x(t_0).$$

7.1 Defining properties

The state-transition matrix satisfies

$$\Phi(t_0, t_0) = I$$

and the composition property

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0).$$

Consequently,

$$\Phi^{-1}(t, t_0) = \Phi(t_0, t).$$

Its derivatives with respect to the first and second time arguments are

$$\frac{\partial}{\partial t}\Phi(t, t_0) = A(t)\Phi(t, t_0),$$

$$\frac{\partial}{\partial t_0}\Phi(t, t_0) = -\Phi(t, t_0)A(t_0).$$

7.2 Worked LTV example

Consider the homogeneous time-varying system

$$\dot{x} = A(t)x, \quad A(t) = \begin{bmatrix} 0 & 0 \\ t & 0 \end{bmatrix}.$$

Its component equations are

$$\dot{x}_1 = 0, \quad \dot{x}_2 = tx_1.$$

Because x_1 is constant, integration gives

$$x_1(t) = c_1, \quad x_2(t) = \frac{1}{2}c_1t^2 + c_2.$$

Two linearly independent solutions are

$$\psi_1(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \psi_2(t) = \begin{bmatrix} 2 \\ t^2 \end{bmatrix}.$$

Placing them in the columns produces the fundamental matrix

$$\Psi(t) = \begin{bmatrix} 0 & 2 \\ 1 & t^2 \end{bmatrix}.$$

Its determinant is -2 , so it is nonsingular for every t . At the initial time t_0 ,

$$\Psi^{-1}(t_0) = \begin{bmatrix} -\frac{t_0^2}{2} & 1 \\ \frac{1}{2} & 0 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \Phi(t, t_0) &= \Psi(t)\Psi^{-1}(t_0) \\ &= \begin{bmatrix} 0 & 2 \\ 1 & t^2 \end{bmatrix} \begin{bmatrix} -\frac{t_0^2}{2} & 1 \\ \frac{1}{2} & 0 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 1 & 0 \\ \frac{t^2 - t_0^2}{2} & 1 \end{bmatrix}}. \end{aligned}$$

The result satisfies the identity property,

$$\Phi(t_0, t_0) = I,$$

and its time derivative is

$$\frac{\partial}{\partial t}\Phi(t, t_0) = \begin{bmatrix} 0 & 0 \\ t & 0 \end{bmatrix} = A(t)\Phi(t, t_0).$$

The lower-left entry also makes composition transparent:

$$\frac{t_2^2 - t_0^2}{2} = \frac{t_2^2 - t_1^2}{2} + \frac{t_1^2 - t_0^2}{2},$$

which implies

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0).$$

7.3 Forced LTV solution

For

$$\dot{x} = A(t)x + B(t)u(t),$$

variation of constants gives

$$x(t) = \Phi(t, t_0)x_0 + \int_{t_0}^t \Phi(t, \tau)B(\tau)u(\tau) d\tau.$$

This is the same initial-condition-plus-input structure as the LTI formula.

7.4 LTI specialization

When $A(t) = A$ is constant,

$$\Phi(t, t_0) = e^{A(t-t_0)}.$$

The dependence reduces from two separate times to the elapsed time $t - t_0$. This time-shift property is a consequence of time invariance.

8 Constant-input example

For a nonsingular A and constant input $u(t) = u_0$,

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu_0 d\tau.$$

Changing variables and integrating gives

$$x(t) = e^{At}x_0 + A^{-1}(e^{At} - I)Bu_0.$$

The equilibrium associated with u_0 is

$$x^* = -A^{-1}Bu_0,$$

so the solution can be written compactly as

$$x(t) = x^* + e^{At}(x_0 - x^*).$$

This expression says that the displacement from equilibrium evolves according to the homogeneous dynamics.

9 Common mistakes

1. **Dropping the initial condition in the Laplace transform.**

The correct identity is $\mathcal{L}\{\dot{x}\} = sX(s) - x(0)$.

2. **Reversing the kernel in the forced response.**

The input applied at time τ propagates to time t through $e^{A(t-\tau)}$ or $\Phi(t, \tau)$.

3. **Treating $e^{A(t-\tau)}$ as a scalar exponential.**

Matrix multiplication order matters.

4. **Assuming $e^{A+B} = e^A e^B$ without checking commutativity.**

5. **Calling every fundamental matrix the state-transition matrix.**

A fundamental matrix is not unique. The normalized combination $\Psi(t)\Psi^{-1}(t_0)$ is the unique state-transition matrix.

6. **Using $e^{\int A(t) dt}$ for a general LTV system.**

This shortcut is generally invalid when $A(t_1)$ and $A(t_2)$ do not commute.

10 Concept checks

1. Why does $e^{A(t-t_0)}x_0$ satisfy the initial condition?
 2. Which part of the response remains when $u = 0$?
 3. Which part remains when $x(t_0) = 0$?
 4. Why does the resolvent become singular at an eigenvalue of A ?
 5. If $\Psi(t)$ is a fundamental matrix, why is $\Psi(t)M$ also fundamental for every constant nonsingular M ?
 6. Use composition to show that $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$.
 7. What additional condition permits $e^{(A+B)t} = e^{At}e^{Bt}$?
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11 Computational companion

The Week 3 notebook illustrates:

- homogeneous motion computed with `scipy.linalg.expm` and `solve_ivp`;

- zero-input and zero-state decomposition;
- forced response computed by numerical integration, convolution, and **scipy.signal.lsim**;
- identity, inverse, and composition properties of state-transition matrices;
- numerical construction of an LTV state-transition matrix; and
- transient amplification in a stable non-normal system.

12 Summary

For an LTI system,

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau) d\tau.$$

For an LTV system,

$$x(t) = \Phi(t, t_0)x_0 + \int_{t_0}^t \Phi(t, \tau)B(\tau)u(\tau) d\tau.$$

The matrix exponential is the LTI state-transition matrix, and its Laplace transform is the resolvent $(sI - A)^{-1}$. These formulas separate initial-condition effects from input effects and provide the foundation for controllability, observability, feedback, and estimation.